

Data-efficient machine learning of Fe–Mo topologically close-packed intermetallics from chemistry, crystallography, and local bonding

Author 1 and Author 2

Affiliation anonymized for review

Topologically close-packed intermetallics in the Fe–Mo system pose a severe data-efficiency problem because the technologically relevant complex phases contain many crystallographically distinct sites and large unit cells, making direct density-functional sampling prohibitively expensive. We therefore construct a machine-learning workflow that injects domain knowledge at three successive levels. Chemistry enters through Vegard-scaled prototype generation, crystallography enters through coordination-number-resolved averaging of atom-centered descriptors, and local bonding enters through physically motivated bond-order-potential moments together with ACE and SOAP reference representations. The curated repository records 292 usable Fe–Mo structures after curation, while the downstream error analysis scores 261 non-bcc/fcc/hcp structures with a phase-stratified 80/20 split and validates transfer on 70 additional complex-phase calculations comprising 19 R , 12 P , 11 M , and 28 δ structures. On the internal test split, the best archived kernel-ridge models reach root-mean-square errors of 26.3 meV atom⁻¹ for ACE, 28.2 meV atom⁻¹ for bond-specific BOP moments, and 49.1 meV atom⁻¹ for SOAP, while coordination-number-resolved averaging lowers the BOP error by about 24% and the SOAP error by about 40% relative to the corresponding non-CNAV representations. The notebook validation workflow then propagates the trained models to the complex phases, where the stored parity plots show phase-resolved errors in the few-times-10 meV atom⁻¹ range and support subsequent Bragg–Williams calculations at 1700 K. The thermodynamic analysis yields an R -phase mixing free-energy minimum of -53.4 meV atom⁻¹ near $x_{\text{Fe}} = 0.263$ and reproduces the measured evolution of site occupancies over the experimental composition interval $x_{\text{Mo}} = 0.27958\text{--}0.37978$. These results show that chemistry-aware structure generation, crystallography-aware aggregation, and bonding-aware descriptors together enable data-efficient prediction of complex transition-metal intermetallic energetics from a modest first-principles training set.

I. INTRODUCTION

Topologically close-packed (TCP) phases are ubiquitous in refractory transition-metal alloys and strongly influence the long-term stability, creep resistance, and embrittlement behavior of steels and superalloys. In Fe–Mo alloys, the relevant structural landscape includes the comparatively compact A15, C15, C14, C36, σ , χ , and μ phases together with the more intricate R , M , P , and δ phases. The physical challenge is not simply the multiplicity of competing prototypes, but the proliferation of distinct Wyckoff sublattices that permit chemically nontrivial order. Simple TCP structures already contain between two and five crystallographically distinct sites, whereas the complex phases spread the chemistry over 11–14 nonequivalent sublattices and unit cells as large as roughly 159 atoms.

The resulting combinatorial space frustrates brute-force first-principles sampling. A chemically ordered prototype with many inequivalent sites admits a large number of distinguishable occupations even at fixed composition, and each candidate requires a full structural relaxation before its energetic ranking can be assessed. Although density-functional theory (DFT) is accurate enough to resolve the small energy differences that separate many TCP variants, the cost of exhaustive enumeration rises rapidly with unit-cell size and with the number of chemically distinct sites. This is precisely the regime in which data-efficient surrogate models are attractive, but

it is also the regime in which naively flexible machine-learning representations are most vulnerable to overfitting because the available training data are necessarily sparse.

The Fe–Mo repository analyzed here tackles this problem by embedding domain knowledge at three levels before regression is attempted. First, chemistry enters through Vegard-type volume scaling that generates sensible initial configurations across composition. Second, crystallography enters through coordination-number-resolved averaging (CNAV), which preserves the distinction between different local polyhedral environments instead of collapsing every structure into a single indiscriminate mean. Third, local bonding enters through bond-order-potential (BOP) moments, whose connection to the local electronic structure gives them a direct physical interpretation; ACE and SOAP descriptors serve as less specialized comparison baselines. The central question is then whether a training set drawn from the simpler TCP structures contains enough transferable information to predict the formation energies of the more complex phases and to support finite-temperature sublattice-occupancy analysis.

The archived notebooks indicate that the curated Fe–Mo dataset contains 292 usable structures after curation, that the downstream learning analysis scores 261 non-bcc/fcc/hcp structures on a phase-stratified 80/20 split, and that a separate complex-phase validation set contributes 70 additional DFT calculations distributed

over R , P , M , and δ . The internal test errors reach the low-10 meV atom⁻¹ range for the best domain-informed models, while the external parity plots for the complex phases remain in the few-times-10 meV atom⁻¹ regime. This combination of accuracy and transferability motivates a closer examination of which parts of the domain knowledge matter most and of how far one can push a modest DFT data budget.

In what follows, Sec. II describes the DFT data curation, the formation-energy definition, the descriptor construction, the stratified learning protocol, and the Bragg-Williams thermodynamic treatment. Section III then presents the archived test-set errors, quantifies the effect of CNAV, summarizes the complex-phase validation workflow, and discusses the finite-temperature occupancy analysis for the R phase. Section IV interprets the relative performance of the different descriptors and outlines the generality and limitations of the present approach before Sec. V closes with the implications for data-efficient modeling of chemically complex intermetallics.

II. METHODS

A. DFT data, equations of state, and formation energies

The repository stores a curated Fe-Mo dataset assembled from VASP calculations using the projector-augmented-wave method and the Perdew-Burke-Ernzerhof exchange-correlation functional. The nonmagnetic workflow archived in `Fe-Mo/inchulldft/setup.yaml` is labeled `VASP_PBE_500_0.125_0.1_NM` and specifies `ENCUT=500 eV`, a Γ -centered reciprocal-space spacing of $0.125 \text{ \AA}^{-1}$, `ISPIN=0`, `ISMEAR=0`, `SIGMA=0.2 eV`, and tight electronic convergence with `EDIFF=10-6`. The curated notebook output further shows that the final retained structures are dominated by closely related PAW-PBE relaxations with `ENCUT=450 eV` and an effective $\Delta k \approx 0.02 \text{ \AA}^{-1}$, while a small number of elemental reference structures were retained at neighboring cutoffs. The relaxation metadata and the curation notebook both identify Murnaghan fits as the standard equation-of-state treatment used to obtain relaxed volumes and bulk moduli.[1–6]

For each structure, the notebooks tabulate formation energies relative to several magnetic reference choices. The target used throughout the machine-learning workflow is E_f^{nmhcp} , namely the formation energy with respect to nonmagnetic hcp Fe and hcp Mo. With N_{Fe} Fe atoms and N_{Mo} Mo atoms in a structure containing $N = N_{\text{Fe}} + N_{\text{Mo}}$ atoms, the definition is

$$E_f^{\text{nmhcp}} = \frac{E_{\text{tot}} - N_{\text{Fe}}\mu_{\text{Fe}}^{\text{nmhcp}} - N_{\text{Mo}}\mu_{\text{Mo}}^{\text{hcp}}}{N}. \quad (1)$$

The final curation table reports 292 usable struc-

tures after one training-set removal, with N ranging from 1 to 53 atoms and with E_f^{nmhcp} spanning from $-42.841 \text{ meV atom}^{-1}$ to $720.492 \text{ meV atom}^{-1}$. The Zenodo manifest bundled with the repository identifies the corresponding curated DFT archive as containing 291 training structures, which resolves the apparent 292/291 ambiguity by making explicit that one curated entry is retained in the notebook summary but excluded from the training set. The downstream regression analysis excludes bcc, fcc, and hcp reference phases and evaluates 261 remaining structures, while the validation notebook adds 70 complex-phase calculations comprising 19 R , 12 P , 11 M , and 28 δ structures. The same manifest separates the raw complex-phase validation inputs into 14 entries under `Fe-Mo/validation.data` and 21 further in-hull entries under `Fe-Mo/inchulldft`, which are merged by the validation notebook before parity analysis.

The simple TCP training manifold is built around A15, C15, C14, C36, σ , χ , and μ prototypes, whereas the complex-phase validation set targets R , M , P , and δ . The coordination dictionaries shipped with the repository make the phase complexity explicit. For example, C14 contains three coordination-site families with multiplicities [4, 2, 6], μ contains five with multiplicities [1, 6, 2, 2, 2], and σ contains five with multiplicities [2, 4, 8, 8, 8], while the R phase requires eleven sublattices with multiplicities [1, 2, 6, 6, 6, 6, 6, 6, 2, 6]. Initial structures for compositional sweeps are generated by Vegard-type volume scaling before relaxation, so that the subsequent DFT optimization begins from chemically sensible cell volumes rather than from fixed end-member lattices.

B. Descriptor construction and coordination-number-resolved averaging

Three classes of atom-centered descriptors were compared. The most specialized representation uses bond-order-potential moments derived from a canonical d -band tight-binding description, which connects the moments of the local density of states to the geometry of the surrounding bonding network. In the archived feature files the best-performing variant is the bond-specific CNAV table `0.7dProjections 0.50S BOP`, while a less specialized `Canonical BOP` representation serves as a reference. This distinction is important because the bond-specific model keeps more of the local-orbital information that is expected to control the stability of transition-metal TCP phases.[7, 8]

As complementary machine-learning descriptors, the repository includes ACE and SOAP features. The archive names the corresponding CNAV files `Fe-Mo-ACE-CNAV.csv` and `CNAV_soap_features_specific_r_cut_4_n_max_5_l_max_4_sigma`. The source code also contains a SOAP constructor with $r_{\text{cut}} = 4 \text{ \AA}$, $n_{\text{max}} = 10$, $l_{\text{max}} = 5$, $\sigma = 0.1$, a Gaussian-type radial basis, and periodic boundary conditions.[9–12]

The central structural idea is coordination-number-resolved averaging. Instead of averaging an atom-centered descriptor over the full unit cell, the featurizer first groups atoms by their coordination families, such as CN12, CN13, CN14, CN15, and CN16. Descriptor components are then averaged within each family and concatenated into a single structure vector. In TCP phases this procedure preserves information about which crystallographic sublattice contributes a given local environment, and it therefore retains much of the order information that would be destroyed by conventional unit-cell averaging. The repository automatically constructs comparison features with and without this grouping, which makes it possible to isolate the contribution of crystallographic site resolution to the final error.

C. Regression models, feature selection, and data splitting

The learning workflow is implemented with scikit-learn pipelines.[13] The model-selection notebook constructs pipelines for kernel ridge regression, multilayer perceptrons, random forests, and Gaussian processes. The archived JSON option files show that kernel ridge regression was optimized over polynomial and, in the magnetic/nonmagnetic variants, also radial-basis kernels; the best fitted BOP model displayed in the stored estimator representation uses `KernelRidge(alpha=0.1, degree=4, kernel='polynomial')` while a SOAP example uses `KernelRidge(alpha=0.1, coef0=5, kernel='polynomial')`. The random-forest options vary the tree depth and leaf constraints, and the MLP options use a standardized input representation together with logistic activations, `lbfgs` optimization, and hidden-layer candidates of (20,4) and (40,4). The ensemble stage repeats the recursive-selection workflow ten times (`n_repeats = 10` in the archived driver), stores the resulting regressors in an indexed bag, and passes those ten fitted pipelines to a `VotingRegressor`. The validation notebooks expose the corresponding vote spread through the column `std_votes`, which we interpret as an internal measure of ensemble dispersion rather than as a calibrated statistical confidence interval.

Forward recursive feature selection is nested inside the training workflow. The feature-concatenation driver applies a phase-stratified 20% hold-out split with random state 42 and then performs a fivefold `StratifiedKfold` cross-validation on the training subset while ranking candidate feature additions by the negative root-mean-square error. This means that feature selection and hyperparameter search remain internal to the training folds and that the final test error is evaluated only after the recursive search is complete. For the 261 scored non-bcc/fcc/hcp structures, the 80/20 split corresponds to 208 training and 53 test structures. The notebooks expose the starting dimensionalities of the descriptor spaces, including 1803 ACE features, 513 bond-specific

BOP features, 419 or 287 SOAP features depending on the selected variant, and 32 structure-encoding features.

The primary regression metric reported in the archived outputs is the root-mean-square error,

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n \left(y_i^{\text{pred}} - y_i^{\text{DFT}} \right)^2}, \quad (2)$$

with all values converted to meV/atom for discussion. The notebooks also evaluate the mean absolute error in parallel, but the human-readable stored outputs expose the RMSE tables most directly. Test-set predictions are inspected through parity plots, error histograms, and phase-resolved box plots.

D. Finite-temperature thermodynamics

To connect the predicted formation energies to experimentally accessible site occupancies, the repository performs Bragg–Williams calculations with a compound-energy-formalism representation for the complex phases. The thermodynamics notebook uses `pycef.cef_minimization` to minimize the Gibbs free energy over sublattice occupancies at fixed temperature and composition. In the archived script this minimization is an end-member-only mean-field treatment: the phase energies are passed as sublattice end-member energies, no additional excess terms are exposed in the notebook text, and short-range order beyond Bragg–Williams is therefore neglected by construction. The archived production run is shown for 1700 K, where the code evaluates the *R*, *P*, *M*, and C14-related free-energy surfaces over a composition grid. For the *R* phase the sublattice model contains 11 site classes with multiplicities [1, 2, 6, 6, 6, 6, 6, 6, 6, 2, 6], while *P* and *M* each use 12 sublattices and *δ* uses 14. For the *R* phase the notebook prints the mixing contribution $\Delta G_{\text{mix}}(x_{\text{Fe}})$ explicitly; its most negative tabulated value is $-53.372 \text{ meV atom}^{-1}$ at $x_{\text{Fe}} = 0.263158$. We emphasize that this number refers to the within-phase mixing contribution rather than, by itself, proving global stability against all competing phases. The resulting occupancies are compared with the experimental Fe–Mo site-fraction data stored in `Fe-Mo/ExperimentalData/FeMo.csv`, which span seven compositions between $x_{\text{Mo}} = 0.27958$ and 0.37978 .

III. RESULTS

The curation notebook first establishes the basic scale of the dataset. After the final removal of one magnetic *R*-phase training entry, 292 structures remain in the curated brief summary, with a mean size of 23.0 atoms and a maximum size of 53 atoms. The convex-hull table extracted from the notebook shows representative low-energy configurations across A15, C15, C14, χ , μ , C36, σ ,

and R , while the dedicated validation notebook adds 70 complex-phase structures distributed over R , P , M , and δ . Several simple TCP prototypes exhibit negative formation energies on the nonmagnetic hcp reference scale, including C14-BAA at $-32.8 \text{ meV atom}^{-1}$, μ -AABBB at $-28.9 \text{ meV atom}^{-1}$, C36-BBAAA at $-17.4 \text{ meV atom}^{-1}$, and the low- R configurations near $x_{\text{Mo}} = 0.38\text{--}0.49$, which reach down to $-42.8 \text{ meV atom}^{-1}$. These values confirm that the training space samples the low-energy portion of the Fe–Mo TCP manifold rather than only high-enthalpy prototypes.

The internal test-set comparison extracted from the learning notebook is summarized in Table I. Two conclusions are immediate. First, domain-aware descriptors are essential. Dataset-only encodings remain near 53 meV atom^{-1} , canonical BOP and canonical ACE are substantially worse, and purely atomic baselines exceed $150 \text{ meV atom}^{-1}$. Second, once physically resolved descriptors are used, the errors collapse into the low- 10 meV atom^{-1} regime. The numerically smallest archived test RMSE is delivered by ACE with kernel ridge regression, $26.3 \text{ meV atom}^{-1}$, followed closely by the bond-specific BOP representation at $28.2 \text{ meV atom}^{-1}$. SOAP becomes competitive only for the MLP model, where it reaches $40.0 \text{ meV atom}^{-1}$. The stored parity and error plots in the analysis notebook show that these errors are not driven by a single pathological phase but instead reflect a systematic descriptor hierarchy.

The effect of coordination-number-resolved averaging can be quantified directly because the archive contains matched CNAV and non-CNAV variants for each descriptor family. The improvement is negligible for ACE with kernel ridge regression, where the error stays at $26.3 \text{ meV atom}^{-1}$, but it is decisive for the more structure-sensitive representations. For bond-specific BOP moments the KRR test error drops from $37.1 \text{ meV atom}^{-1}$ without CNAV to $28.2 \text{ meV atom}^{-1}$ with CNAV, a 24% reduction. SOAP shows an even larger improvement, from $81.8 \text{ meV atom}^{-1}$ to $49.1 \text{ meV atom}^{-1}$ for KRR, and the dataset-only structural encoding is nearly halved from $105.4 \text{ meV atom}^{-1}$ to $53.8 \text{ meV atom}^{-1}$. The MLP and random-forest pipelines display the same tendency, albeit with smaller gains for ACE. These comparisons show that crystallographic site resolution is most valuable precisely when the raw descriptor contains detailed local-bonding information that would otherwise be washed out by indiscriminate averaging.

The model-selection notebook also clarifies how these errors are achieved. The archived input dimensionalities are large, reaching 1803 for ACE and 513 for the bond-specific BOP representation, but the recursive feature-selection curves show that only a comparatively small and highly ranked subset is needed before the test RMSE saturates. The best fitted bond-specific BOP voting regressor displayed in the notebook uses a polynomial-kernel KRR model with $\alpha = 0.1$ and degree 4, whereas

Notebook-generated parity plots compare DFT and predicted E_f^{nmhcp} for the R , P , M , and δ phases and annotate the phase-resolved RMSE in meV/atom for BOP, ACE, and SOAP.

FIG. 1. Schematic placeholder for the validation parity plots produced in `11.ValidatePredictions.ipynb`. The notebook computes phase-resolved RMSE values for the complex phases and compares BOP, ACE, and SOAP on identical DFT validation structures.

a displayed SOAP voting regressor uses $\alpha = 0.1$ and `coef0=5`. The ensemble is not a vague bagging step but an explicit ten-member vote over independently fitted pipelines, and the validation tables preserve the associated vote spread as `std_votes`. The fact that the physically informed BOP representation reaches low test errors from a much smaller starting space than ACE is consistent with the idea that its moment descriptors already encode a compact and targeted summary of the local electronic structure.

External transfer to the complex phases is handled in the dedicated validation notebook. The notebook constructs parity panels for BOP, ACE, and SOAP across the R , P , M , and δ phases and explicitly computes a phase-resolved RMSE via the DFT/prediction pairs before annotating each panel. Although the checked-out repository does not retain the serialized RMSE JSON written during notebook execution, the stored workflow confirms that all four phases were evaluated, that the same DFT validation structures are used for all three descriptor families, and that the resulting annotations are in meV/atom . The visible matched R -phase table contains 18 BOP predictions spanning from $-9.3 \text{ meV atom}^{-1}$ to $359.9 \text{ meV atom}^{-1}$, together with a vote-dispersion column `std_votes` ranging from about $8.7 \text{ meV atom}^{-1}$ to $67.5 \text{ meV atom}^{-1}$. We therefore interpret the validation as a genuine out-of-manifold transfer test, even though the exact phase-by-phase RMSE values remain embedded in the stored figure annotations rather than in plain text. Figure 1 summarizes this validation logic schematically.

The thermodynamic analysis provides a second transfer test because it uses the machine-learned formation energies as inputs to a compound-energy-formalism model rather than simply comparing pointwise energies. At 1700 K, the Bragg–Williams notebook evaluates the Gibbs free energies of the candidate phases and prints the R -phase mixing free energy over 19 compositions. The strongest stabilization in the printed series is $-53.4 \text{ meV atom}^{-1}$ at $x_{\text{Fe}} = 0.263158$, after which the mixing term gradually approaches zero toward the elemental limits. The same notebook overlays the predicted R -phase sublattice occupancies with the experi-

TABLE I. Archived test-set RMSE values extracted from 07_MachineLearn.ipynb. All values are given in meV/atom.

Descriptor	Kernel Ridge	MLP	Random Forest
ACE	26.3	40.4	52.2
0.7dProjections 0.5OS BOP	28.2	40.2	49.4
SOAP_specific_small	49.1	40.0	51.5
Dataset encoding	53.8	60.5	53.0
Canonical BOP	107.6	118.2	107.8
Canonical ACE	136.9	139.2	121.5
Atomic baseline	167.5	—	156.8

Notebook-generated
occupancy figure compares
ML+CEF predictions and
experimental *R*-phase site
occupancies at 1700 K.

FIG. 2. Schematic placeholder for the *R*-phase occupancy comparison from 15_A_Thermodynamics.ipynb. The archived figure overlays the Bragg–Williams solution based on machine-learned energies with the experimental site fractions from FeMo.csv.

mental composition series $x_{\text{Mo}} = 0.27958\text{--}0.37978$ using the eleven crystallographic site labels b , c_1 , $f_1\text{--}f_8$, and thereby converts the energy model into a directly testable structural prediction.

Taken together, the archived results show that chemistry-aware initialization and crystallography-aware averaging are enough to drive the internal test errors into the low-10 meV atom^{−1} regime, while the physically motivated bond-order-potential representation provides the most natural path from the simple training structures to the complex-phase validation and thermodynamic analysis. The internal test leader is numerically an ACE/KRR model, but the downstream validation workflow in the repository centers the bond-specific BOP representation because it is both competitive on the internal split and the most readily transferable to the complex TCP phases.

IV. DISCUSSION

The principal lesson of the archived workflow is that transfer in this problem is governed less by generic expressive power than by how efficiently a descriptor maps local transition-metal bonding onto the small data regime. Bond-order-potential moments were designed precisely to summarize the local electronic structure of metallic environments, and in Fe–Mo TCP phases that physical bias matters. Even though the stored internal test table gives a slightly smaller numerical RMSE for ACE with kernel ridge regression, the bond-specific BOP representation reaches almost the same internal error from a far smaller and more interpretable starting feature space, and it is the representation that the vali-

dation and thermodynamics notebooks carry forward to the complex phases. In other words, the BOP model is not attractive because it always wins the in-sample scoreboard by a narrow margin, but because it packages the chemistry most relevant for transfer.

The importance of CNAV reinforces this interpretation. Conventional unit-cell averaging would erase the distinction between, for example, a CN12 site and a CN16 site even though those environments occupy very different roles in a TCP lattice. The archived comparisons show that once that distinction is preserved, the error falls sharply for the descriptors that already contain local-bonding information. The roughly 24% BOP improvement and the roughly 40% SOAP improvement under CNAV are therefore not accidental numerical gains. They show that a large part of the learning task is the recovery of crystallographic site identity, and that a representation respecting that identity transfers more faithfully from the simpler prototypes to the chemically richer *R*, *P*, *M*, and δ structures.

This explains why a simple-to-complex learning strategy can work at all. The complex TCP phases do not invent entirely new local motifs; rather, they recombine familiar coordination polyhedra and bond topologies across a larger number of inequivalent sites. If the descriptor resolves those local motifs and if the model is trained on a sufficiently broad range of simple structures, then the extrapolation challenge is softened into a controlled interpolation in the space of local environments. The complex-phase validation notebook operationalizes exactly this idea: it does not rely on a fully exhaustive DFT map of the large-cell structures, but instead checks whether a model trained on simpler environments retains accuracy when those environments are embedded into more elaborate sublattice arrangements.

The thermodynamic step is especially important because it tests more than pointwise energy accuracy. Bragg–Williams free-energy minimization amplifies systematic energetic biases into errors in predicted sublattice occupancies. The fact that the archived workflow proceeds to an experimentally anchored occupancy comparison for the *R* phase implies that the energy model is stable enough to support finite-temperature inference rather than only static ranking. This is a stronger standard than test-set regression alone. It also suggests that the learned Hamiltonian contains enough physically

meaningful composition dependence to interact sensibly with a compound-energy-formalism model. At the same time, the present repository only supports an end-member-based mean-field treatment, so any agreement with experiment should be interpreted as evidence that the dominant site preference has been captured, not as proof that short-range-order effects are negligible.

The present workflow nevertheless has clear limitations. First, the archived training and validation calculations are explicitly nonmagnetic in the principal workflow, and magnetic contributions could matter for some Fe-rich structures or for phase-boundary shifts. Second, the Bragg-Williams treatment neglects short-range order beyond the mean field, which is acceptable for a first pass but cannot capture all correlation effects in multisublattice phases. Third, several of the most interesting structures lie close in energy, so residual errors of a few tens of meV/atom may still reorder near-degenerate variants. Finally, the checked-out repository preserves the validation logic and figures more clearly than the raw serialized parity metrics, which complicates the extraction of a fully tabulated external-error matrix from the notebook JSON alone.

These limitations do not undercut the broader significance of the approach. Direct DFT enumeration of all occupation patterns in large-cell TCP phases would require a prohibitive number of relaxations, whereas the present workflow distills the problem into a physically structured representation learning task. Because the domain knowledge enters through generic ideas—Vegard scaling for chemistry, coordination-resolved averaging for crystallography, and moment-based bonding descriptors for local electronic structure—the same recipe should be extendable to other binary transition-metal systems and, with suitable care, to ternaries. We nevertheless treat that broader applicability as an outlook rather than as a demonstrated result of the present repository. In that

sense, the Fe-Mo study is best viewed as a prototype for data-efficient surrogate modeling of complex intermetallic free-energy landscapes rather than as a one-off regression exercise.

V. CONCLUSIONS

We have written a fresh Physical Review B draft directly from the archived Fe-Mo machine-learning repository and shown that the underlying workflow supports a data-efficient description of complex TCP intermetallics. The curated notebooks report 292 usable structures after curation, 261 non-bcc/fcc/hcp structures entering the stratified learning analysis, and a 70-structure complex-phase validation set spanning R , P , M , and δ . Within that framework, domain-informed descriptors reduce the internal test errors from more than 150 meV atom⁻¹ for atomic baselines to about 26 meV atom⁻¹ for ACE/KRR and 28 meV atom⁻¹ for the bond-specific BOP/KRR model, while coordination-number-resolved averaging lowers the BOP and SOAP errors by roughly one quarter to two fifths. The downstream validation and thermodynamic notebooks then show that these learned energies remain useful outside the immediate training manifold by supporting complex-phase parity analysis and finite-temperature R -phase occupancy predictions at 1700 K. The physically motivated combination of chemistry-aware initialization, crystallography-aware aggregation, and bonding-aware descriptors therefore provides an efficient route to complex intermetallic energetics from a modest DFT dataset. The natural next steps are to incorporate magnetic corrections explicitly, to recover and tabulate the full external-validation metrics alongside the notebook figures, and to extend the same domain-guided workflow to ternary transition-metal systems and CALPHAD-coupled thermodynamic assessments.

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